

✓ PYTHON DSA — 3 MONTH (90-DAY) ROADMAP

Daily time: ~1.5 hours

Rule: Move forward even if it feels imperfect

● MONTH 1 — Foundations & Linear DSA (Days 1–30)

◆ ARRAYS & STRINGS (Days 1–10)

~~Day 1:~~

- Array basics, traversal, indexing
- Time & space complexity
(No problems)

~~Day 2:~~

- Prefix sum
- Frequency array
(No problems)

~~Day 3:~~

- Two pointers pattern
(1 easy problem)

~~Day 4:~~

- Sliding window (fixed window)
(1 easy problem)

~~Day 5:~~

- Sliding window (variable window)
(1 medium problem)

~~Day 6:~~

- Kadane's Algorithm
(1 problem)

~~Day 7:~~

- Array rotation, reverse, sorting basics
(2 easy problems)

~~Day 8:~~

- String basics
- Character frequency
(1 problem)

~~Day 9:~~

- String patterns (palindrome, anagram) (2 problems)
- Day 10:**
- REVISION day (No new problems)
- ☞ Stop arrays here. Do NOT restart later.

~~Day 1~~ – HashMap Basics (Very Light)

What is hashing?

Dictionary operations

Frequency counting

Why O(1) lookup matters

Goal: Comfort with dict



◆ HASHING (Days 11–14)

~~Day 11:~~

- Dictionary / Counter
- Frequency patterns (No problems)

Two Sum

Count pairs with sum K

Difference pair problems

Goal: Understand complement logic
target - current

~~Day 3~~ – Prefix Sum Concept Only

What is prefix sum?

Why it helps in subarrays?

Manual dry runs

No complex problems

Goal: Deep clarity, zero confusion



Day 12:

- Hashing for pairs & subarrays (1 medium)

Day 13:

- Prefix hash (1 medium)

Day 14:

- REVISION (1 mixed problem)

Using prefix + hashmap

Existence check

Goal: Understand
prefix - target

Day 5 – Count Subarrays with Sum K

◆ RECURSION & BACKTRACKING (Days 15–20)

Day 15:

- Recursion basics
- Stack flow (No problems)

Frequency of prefix sums

Why {0:1} is important

Goal: Pattern mastery



K

First occurrence trick

Day 16:

- Subsequence pattern (1 problem)

Length calculation

Goal: Advanced confidence

Day 7 – Mixed Practice Day

Day 17:

- Pick / Not pick (1 problem)

Solve 3–4 mixed problems

Recognize pattern without hints

Goal: Pattern recognition ability

Day 18:

- Permutations
(1 problem)

Day 19:

- Combinations
(1 problem)

Day 20:

- REVISION
(No new problems)
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◆ **SORTING & SEARCHING (Days 21–25)**

Day 21:

- Bubble, selection, insertion
(No problems)

Day 22:

- Merge sort concept
(1 problem)

Day 23:

- Binary search basics
(1 problem)

Day 24:

- First/last occurrence
(1 problem)

Day 25:

- REVISION
(1 problem)
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◆ **LINKED LIST (Days 26–30)**

Day 26:

- LL basics
(No problems)

Day 27:

- Reverse linked list
(1 problem)

Day 28:

- Fast & slow pointer
(1 problem)

Day 29:

- Cycle detection
(1 problem)

Day 30:

- REVISION
(No problems)
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● **MONTH 2 — Core Interview DSA (Days 31–60)**

◆ **STACK & QUEUE (Days 31–36)**

Day 31:

- Stack basics
(No problems)

Day 32:

- Parentheses problems
(1 problem)

Day 33:

- Monotonic stack
(1 problem)

Day 34:

- Queue & deque
(1 problem)

Day 35:

- Sliding window using deque
(1 problem)

Day 36:

- REVISION
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◆ **TREES & BST (Days 37–48)**

Day 37:

- Tree basics
(No problems)

Day 38:

- DFS (pre/in/post)
(1 problem)

Day 39:

- Height-based recursion
(1 problem)

Day 40:

- Diameter of tree
(1 problem)

Day 41:

- Level order traversal
(1 problem)

Day 42:

- Balanced tree
(1 problem)

Day 43:

- BST basics
(No problems)

Day 44:

- BST search & insert
(1 problem)

Day 45:

- BST validation
(1 problem)

Day 46–48:

- REVISION
(2 mixed problems)

◆ **HEAPS (Days 49–52)**

Day 49:

- Heap basics
(No problems)

Day 50:

- Kth largest/smallest
(1 problem)

Day 51:

- Heap + hash
(1 problem)

Day 52:

- REVISION
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◆ **GRAPHS (Days 53–60)**

Day 53:

- Graph basics
(No problems)

Day 54:

- BFS
(1 problem)

Day 55:

- DFS
(1 problem)

Day 56:

- Connected components
(1 problem)

Day 57:

- Cycle detection
(1 problem)

Day 58:

- Topological sort
(1 problem)

Day 59:

- Shortest path (BFS)
(1 problem)

Day 60:

- REVISION
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● **MONTH 3 — Advanced & Confidence (Days 61–90)**

◆ **DYNAMIC PROGRAMMING (Days 61–75)**

Day 61:

- DP intro
(No problems)

Day 62:

- 1D DP
(1 problem)

Day 63:

- Take / Not take
(1 problem)

Day 64:

- Subset sum
(1 problem)

Day 65:

- 2D DP
(1 problem)

Day 66:

- Partition DP
(1 problem)

Day 67–70:

- DP REVISION
(3 problems)

◆ **MIXED PRACTICE (Days 71–85)**

- Each day:
 - 1 random topic
 - 1 problem
 - Re-code from memory



Mon–Thu: DSA + Web Dev

Fri–Sat: DSA + AI/ML

Sun: Light revision + optional Web I